Reg. No. | | | | | | | | | |

Question Paper Code : 12665

B.E / B.TECH DEGREE EXAMINATION, NOV / DEC 2016

Third Semester

B.E – Computer Science and Engineering

15CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations : Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. What is Message passing Multicomputers?
2. Define Straight line sequencing.
3. Represent pictorially the B-Cell used in fast Adders.
4. Why is the Wait-For-Memory- Function- Completed (WMFC) step needed when reading from or writing to main memory?
5. Define Instruction throughput for pipelined processors.
6. Compare R7, R8
   Branch>0 TARGET- Does this instruction has side effect? Justify.
7. Define Double-data-rate SDRAM with an example.
8. What is meant by Cache Replacement Algorithm?
9. List the purpose of various registers available in I/O devices.
10. Compare and contrast spatial and temporal locality of reference.

PART B – (5 × 16 = 80 Marks)

11. a) i. Discuss in detail about the design of Full Adder. (8 Marks)

11. a) ii. Compute 17/7 using non restoring division algorithm (8 Marks)

OR

11. b) i. Define addressing mode. Explain different addressing modes supported by modern processors. (8 Marks)

11. b) ii. Perform Carry Save Addition of summands on multiplicand 101101 and multiplier 111111 and state its advantage. (8 Marks)

12. a) i. Discuss the basic organization of Micro programmed control unit for generating the control signals with suitable diagram. (8 Marks)

12. a) ii. List the sequence of actions needed to execute the following instructions in terms of data path components

Store R5,10(R8), SUB R3, R4, R5 (8 Marks)

OR

12. b) i. At some point in the execution of a program, registers R2, R4, and R5 contain the values 3000, 7500, and 2500, respectively. Show the contents of the Multiplexer and the registers RA, RB, RZ, RY, and R6 during steps 3 to 5 as the instruction

Subtract R2, R4, R5 is fetched and executed, and also during step 1 of the instruction that is fetched next. (8 Marks)

12. b) ii. Discuss briefly various control signals used in data path. (8 Marks)

13. a) i. Assume that 30 percent of the dynamic count of the instructions executed for a program is branch instructions. There are no pipeline stalls due to data dependencies.

A. Determine the execution times for two cases: when 70 percent of the branches are taken, and when 30 percent of the branches are taken.

B. Determine the speedup for one case relative to the other. (8 Marks)

13. a) ii. Explain the techniques for alleviating the pipeline stalls due to data dependencies. (8 Marks)

OR

13. b) i. List out the ways to reduce the stall in the pipeline due to branch instruction. Explain the various branch prediction techniques. (8 Marks)

13. b) ii. 1000 Add R3, R2, #30

1004 Subtract R5, R4, #35

1008 And R6, R4, 0X4B

1012 Add R7, R2, R4

Initially, registers R2 and R4 contain 3500 and 50, respectively. These instructions are executed in a computer that has a five-stage pipeline. The first instruction is fetched in clock cycle 1, and the remaining instructions are fetched in successive cycles.

A. Draw a diagram that represents the flow of the instructions through the pipeline. Describe the operation being performed by each pipeline stage during clock cycles 1 through 8.

B. With reference to data path describe the contents of registers IR, PC, RA, RB, RY, and RZ in the pipeline during cycles 2 to 8. (8 Marks)

14. a) i. A block-set-associative cache consists of a total of 256 blocks divided into 4-block sets. The main memory contains 8192 blocks, each consisting of 128 words.

A. How many bits are there in a main memory address?

B. How many bits are there in each of the TAG, SET and WORD fields?

Also, explain the direct and associative mapping techniques for the above memory specifications. (10 Marks)

14. a) ii. What is a virtual memory? Describe any one virtual memory management technique. (6 Marks)

OR

14. b) i. Explain the use of DMA controllers in a computer system with neat diagram. (6 Marks)

14. b) ii. Suppose that a computer has a processor with two L1 caches, one for instructions and one for data, and an L2 cache. Let 5τ be the access time for the two L1 caches. The miss penalties are approximately 10τ for transferring a block from L2 to L1, and 30τ for transferring a block from the main memory to L2. For the purpose of this problem, assume that the hit rates are the same for instructions and data and that the hit rates in the L1 and L2 caches are 0.75 and 0.50, respectively.

A. What fraction of accesses miss in both the L1 and L2 caches, thus requiring access to the main memory?

B. What is the average access time as seen by the processor?

C. Suppose that the L2 cache has an ideal hit rate of 1. By what factor would this reduce the average memory access time as seen by the processor?

D. Consider the following change to the memory hierarchy. The L2 cache is removed and the size of the L1 caches is increased so that their miss rate is cut in half. What is the average memory access time as seen by the processor in this case? (10 Marks)

15. a) i. Write in detail about various interconnection standards. (10 Marks)

15. a) ii. Discuss about program controlled I/O. (6 Marks)

OR

15. b) i. What are the functions of I/O device interface? Explain about parallel interface and the use of interrupts in handling I/O. (10 Marks)

15. b) ii. What is SCSI controller? Write about the sequence of events that take place when processor sends a command to SCSI controller. (6 Marks)